

Advanced Econometrics – Lab 06

1 Count data models

Exercise 1

Let's model number of surgical errors and medical malpractice per hospital. The data set `surgical_errors.csv` includes:

- *hosp_id* - hospital id
- *errors* - number of errors
- *lnN* - logarithm of operation number
- *private* - a dummy variable = 1 if the hospital is a private firm, 0 = if it is a part of national health service

- a) Estimate Poisson regression model.
- b) Estimate Negative Binomial regression model.
- c) Compare both models.

Exercise 2

Let's analyse total number of medals won by a country at Olympic Games. The data set contains information on number of medal won (*medaltot*), population (*pop*), Gross Domestic Product (*gdp*). What is more, it includes two dummy variables: *soviet* = 1 if a country was part of the Soviet Block, and *host* = 1 for Olympic Games host-countries.

- a) Estimate Poisson regression model.
- b) Estimate Negative Binomial regression model.
- c) Compare both models.
- d) Check frequencies of zero observations.
- e) Estimate Zero-Inflated Poisson regression model.
- f) Estimate Zero-Inflated Negative Binomial regression model.
- g) Compare all the models.

Exercise 3¹

Deb and Trivedi (1997) analyze data on 4406 individuals, aged 66 and over, who are covered by Medicare, a public insurance program. Originally obtained from the US National Medical Expenditure Survey (NMES) for 1987/88, the data are available from the data archive of the Journal of Applied Econometrics at <http://www.econ.queensu.ca/jae/1997-v12.3/deb-trivedi/>. It was prepared for an R package accompanying Kleiber and Zeileis (2008) and is also available as `DebTrivedi.rda` in the Journal of Statistical Software together with Zeileis (2006). The objective is to model the demand for medical

¹Source: <https://cran.r-project.org/web/packages/pscl/vignettes/countreg.pdf>

care|as captured by the number of physician/non-physician office and hospital outpatient visits-by the covariates available for the patients. Here, we adopt the number of physician office visits *ofp* as the dependent variable and use the health status variables *hosp* (number of hospital stays), *health* (self-perceived health status), *numchron* (number of chronic conditions), as well as the socioeconomic variables *gender*, *school* (number of years of education), and *privins* (private insurance indicator) as regressors. For convenience, we select the variables used from the full data set:

```
R> data <- DebTrivedi[, c(1, 6:8, 13, 15, 18)]
```

- a) Estimate Poisson regression model.
- b) Estimate Negative Binomial regression model.
- c) Check frequencies of zero observations.
- d) Estimate Zero-Inflated Poisson regression model.
- e) Estimate Zero-Inflated Negative Binomial regression model.
- f) Estimate the Hurdle model.
- g) Compare all the models.